Arslan Arshad

Software Engineer

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Portfolio

Objective

Full-Stack Software Engineer with expertise in Python (Django REST Framework, FastAPI) and Node.js, specializing in scalable backend systems, RESTful APIs, and database optimization. Skilled in PostgreSQL, MySQL, and NoSQL, with hands-on experience in AI, Generative AI, and Agentic AI including LLM integration and RAG pipelines. Proficient in Git and deployment across Linux/Windows, with additional capability in ReactJS and modern JavaScript frameworks.

Expertise

- **Programming Languages:** Python, Django, FastAPI, ReactJs, JavaScript, LangChain, LangGraph, C++, Java
- **Platforms:** Windows, Linux
- **IDEs:** VS Code, Visual Studio, Brackets, IntelliJ IDEA

Experience

Inovaqo (Full Stack Software Engineer) Jan 2025 – Present

- Designed and developed full-stack applications with a backend focus using Django REST Framework and FastAPI, and ReactJS for interactive, scalable frontends.
- Built and optimized RESTful APIs, database models, and caching layers to ensure secure, high-performance system integration.
- Engineered AI-driven features including LLM-powered chatbots, RAG pipelines, and intelligent search systems by integrating LangChain, LangGraph, and vector databases.
- Implemented Agentic AI workflows with multi-agent orchestration, enabling autonomous task execution and decision support across applications.
- Collaborated with product, AI, and frontend teams to deliver production-ready GenAI solutions with seamless UI/UX and scalable cloud deployment.

Coding Cops (Associate Software Engineer) Sep 2023 – Jan 2025

- Developed and maintained full-stack web applications using Django (REST Framework) for backend and ReactJS for frontend.
- Designed and integrated RESTful APIs to enable efficient communication between frontend and backend systems.
- Implemented state management in React to enhance application performance, scalability, and maintainability.
- Collaborated with cross-functional teams to deliver responsive, interactive, and user-friendly web applications.

Education

- **Punjab University College of Information and Technology (PUCIT)** Lahore, Pakistan
  BS Software Engineering

Projects

AI-Powered Market Research Platform *2025*

Technologies: Python, FastAPI, LangChain, Groq AI, PinoCone, React

- Developed a comprehensive market research application using **React**, **FastAPI**, and **LangChain** with multi-agent AI system
- Implemented **LLaMA** integration via Groq API for intelligent report generation and competitive analysis
- Designed RESTful API architecture with **SQLite** database, **Pydantic** validation, and **Pinecone** vector storage
- Created modular multi-agent system featuring Scout, Synthesizer, Critic, and Reporter agents for automated data processing
- Integrated real-time web search capabilities and template-based report generation for various market analysis types
- Implemented user authentication, analytics dashboard, and export functionality supporting PDF, DOCX, and PPTX formats

YouTube Talker -AI-Powered Video Content Analyst *2025*

Technologies: Python, Flask, LangChain, Google Gemini AI, FAISS, YouTube API

- Developed intelligent web application enabling natural language interaction with YouTube video content through automated transcript extraction and RAG-based question answering
- Implemented RESTful API endpoints using Flask framework with modular service architecture for YouTube content processing and AI-powered responses
- Built advanced RAG pipeline integrating LangChain, Google Gemini AI, and FAISS vector database for semantic search and contextual content understanding
- Designed responsive web interface with real-time chat functionality, typewriter effects, and Bootstrap framework for enhanced user experience
- Optimized text processing with intelligent chunking strategies, embedding models, and configurable similarity search parameters

Genuity ReactJS, Django REST Framework

- Developed a robust project management tool inspired by Jira, enhanced with features for managing both human and physical resources.
- Built dynamic frontend using ReactJS with Ant Design and Material-UI, and scalable APIs with Django REST Framework.

Certificates

- **Introduction to HTML5:** Certificate of completion (Coursera).
- **C for Everyone: Programming Fundamentals:** Certificate of completion (Coursera).
- **Python Data Structures:** Certificate of completion (Coursera).
- **Programming for Everybody (Getting Started with Python):** Certificate of completion (Coursera).

Hobbies and Interests

Table Tennis
TV Series
Snooker
Chess
Cricket
E-Games
Badminton
Sudoku